

HOLODECK

Machines virtuelles Web pour les ingénieurs de Starfleet

Infrastructure serveur Debian multi-services (DHCP · DNS · Web · FTP · LDAP)

Contexte du projet

La Fédération équipe le Holodeck de l'USS Enterprise-D de machines virtuelles pour développer les sites web stellaires.

1

Deux VM Debian

Une VM Serveur (sans GUI) + une VM Client (avec interface graphique) reliées en réseau privé.

2

Serveur multi-services

DHCP, DNS, Web (Nginx/PHP/MariaDB), FTP sécurisé, annuaire LDAP.

3

Sécurité imposée

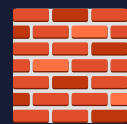
Pas de compte sudo, pare-feu restrictif, versions officielles à jour, HTTPS partout.

Sécurité de l'infrastructure



Pas de compte sudo

Travail en root via su -, aucun paquet sudo installé



Pare-feu restrictif

nftables — policy drop, ports strictement requis : 22, 53, 67, 80, 443, 21 + plage FTP passive



Certificat SSL wildcard

*.starfleet.lan — auto-signé, réutilisé pour Nginx et FTP



Authentication LDAP

Connexion au site web via annuaire OpenLDAP

Architecture réseau

Domaine : starfleet.lan

VM SERVEUR (Debian)

ens33 (WAN) — DHCP automatique

ens34 (LAN) — 192.168.100.1/24

- DHCP (isc-dhcp-server)
- DNS (bind9)
- Nginx — HTTPS, 4 sites
- PHP 7.4 + PHP 8.4 (FPM)
- MariaDB
- vsftpd — FTPS chrooté
- slapd — LDAP



Host-only

VM CLIENTE (Debian + GNOME)

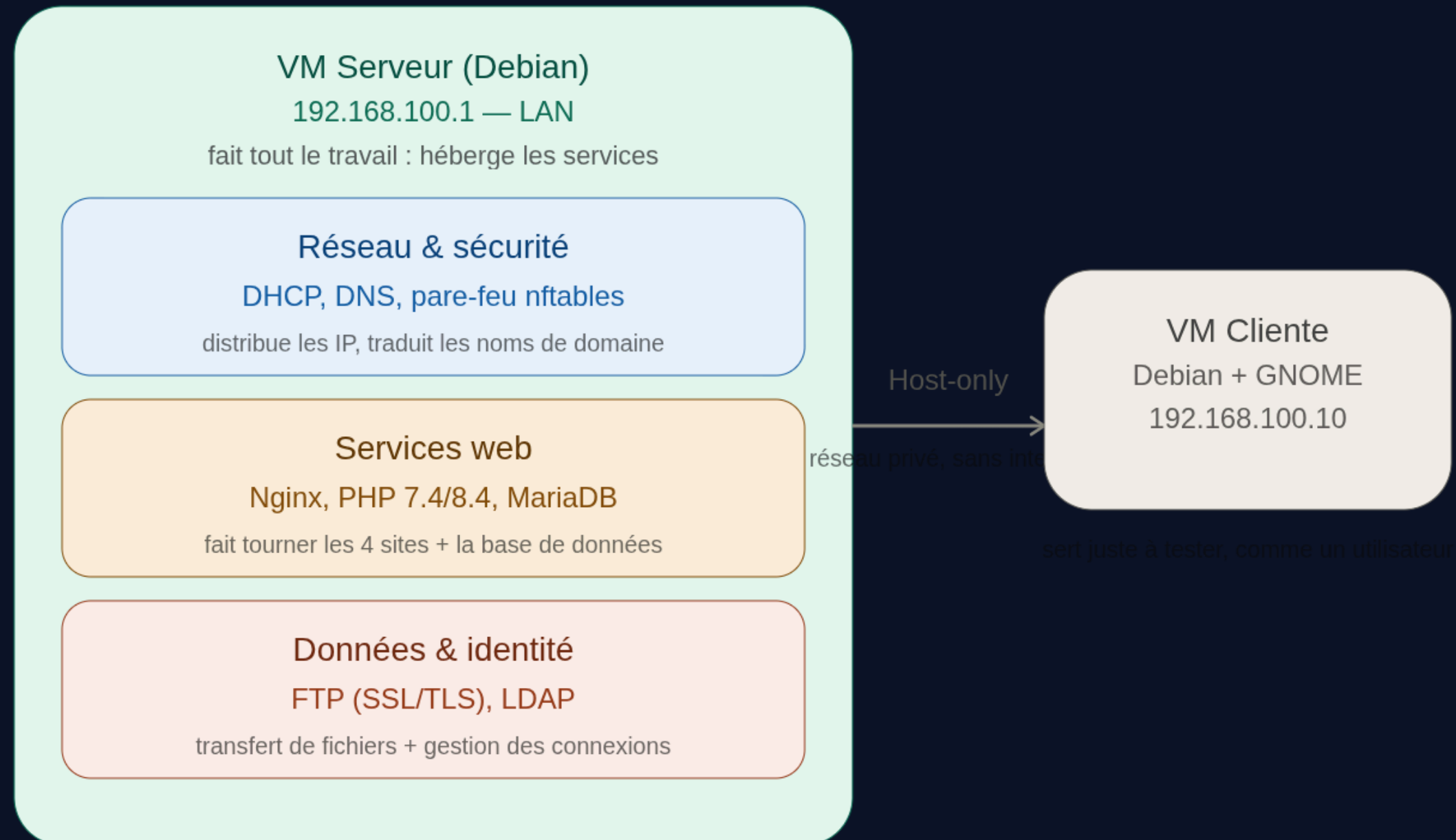
Interface graphique GNOME

IP DHCP : 192.168.100.10

Firefox — tests HTTPS

Fichiers (Nautilus) — test FTPS

Navigateur — test LDAP



Services installés — VM Serveur

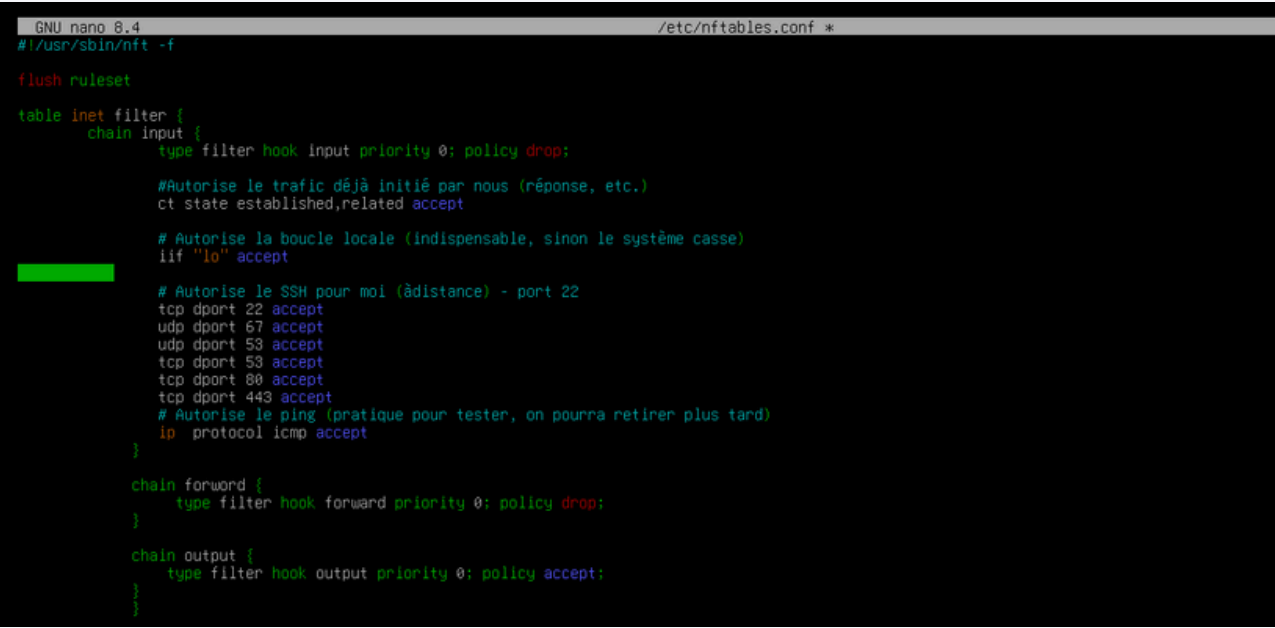
Nginx	1.30.4	Dépôt officiel nginx.org — HTTPS, redirection 80→443
PHP	7.4 + 8.4	Dépôt Sury — cohabitation via PHP-FPM (sockets séparés)
MariaDB	12.3.3	Dépôt officiel — sécurisée via mariadb-secure-installation
DHCP / DNS	isc-dhcp-server / bind9	Domaine starfleet.lan
FTP	vsftpd	SSL/TLS obligatoire, chrooté sur /var/www
LDAP	OpenLDAP (slapd)	dc=starfleet,dc=lan

Services installés — VM Serveur

Nginx	1.30.4	Dépôt officiel nginx.org — HTTPS, redirection 80→443
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```
root@holdeck-serveur:~# curl https://nginx.org/keys/nginx_signing.key | gpg --dearmor | tee /usr/share/keyrings/nginx-archive-keyring.gpg >/dev/null
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 11809  100 11809    0     0  15742      0  --:--:-- --:--:-- --:--:-- 15724
root@holdeck-serveur:~#
```

Ajout de la clé officielle de Nginx

A screenshot of the GNU nano 8.4 text editor showing the configuration of nftables in the file /etc/nftables.conf. The configuration includes a table named 'inet' with a filter 'filter' and a chain 'input'. The 'input' chain has a hook at priority 0 with policy 'drop'. It contains rules for accepting established and related connections, local loopback traffic, SSH on port 22, various ports (67, 53, 80, 443), and ICMP. There are also forward and output chains configured with similar hooks and policies.

```
GNU nano 8.4 /etc/nftables.conf *
#!/usr/sbin/nft -f

flush ruleset

table inet filter {
    chain input {
        type filter hook input priority 0; policy drop;

        # Autorise le trafic déjà initié par nous (réponse, etc.)
        ct state established,related accept

        # Autorise la boucle locale (indispensable, sinon le système casse)
        iif "lo" accept

        # Autorise le SSH pour moi (à distance) - port 22
        tcp dport 22 accept
        udp dport 67 accept
        udp dport 53 accept
        tcp dport 53 accept
        tcp dport 80 accept
        tcp dport 443 accept
        # Autorise le ping (pratique pour tester, on pourra retirer plus tard)
        ip protocol icmp accept
    }

    chain forward {
        type filter hook forward priority 0; policy drop;
    }

    chain output {
        type filter hook output priority 0; policy accept;
    }
}
```

```
root@holdeck-serveur:~# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@holdeck-serveur:~#
```

Nginx sert bien de page Web

Rajout du port 80 et 443 pour une redirection

Services installés — VM Serveur

PHP

7.4 + 8.4

Dépôt Sury — cohabitation via PHP-FPM (sockets séparés)

Ajout du dépôt Sury celui qui permet d'installer les dernières versions officielles de PHP (7.4 et 8.4)

```
root@holdeck-serveur:~# curl -sSL http://packages.sury.org/php/apt.gpg | gpg --dearmor | tee /usr/share/keyrings/sury-php-archive-keyring.gpg >/dev/null
root@holdeck-serveur:~# echo "deb [signed-by=/usr/share/keyrings/sury-php-archive-keyring.gpg] http://packages.sury.org/php/ trixie main" | tee /etc/apt/sources
.list.d/sury-php.list
deb [signed-by=/usr/share/keyrings/sury-php-archive-keyring.gpg] http://packages.sury.org/php/ trixie main
root@holdeck-serveur:~# apt update
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Réception de : 2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Réception de : 3 http://security.debian.org/debian-security trixie-security InRelease [43,4 kB]
Réception de : 5 http://security.debian.org/debian-security trixie-security/main Sources [227 kB]
Atteint : 6 http://nginx.org/packages/debian trixie InRelease
Réception de : 7 http://security.debian.org/debian-security trixie-security/main amd64 Packages [256 kB]
Réception de : 8 http://security.debian.org/debian-security trixie-security/main Translation-en [156 kB]
Réception de : 4 https://packages.sury.org/php trixie InRelease [6 129 B]
Réception de : 9 https://packages.sury.org/php trixie/main amd64 Packages [286 kB]
1 022 ko réceptionnés en 1s (1 090 ko/s)
Tous les paquets sont à jour.
root@holdeck-serveur:~# _
```

```
root@holdeck-serveur:~# ls -la /run/php/
total 0
drwxr-xr-x  2 www-data www-data  80  9 sept. 09:36 .
drwxr-xr-x 22 root      root      640  9 sept. 09:33 ..
srw-rw----  1 www-data www-data   0  9 sept. 09:36 php7.4-fpm.sock
srw-rw----  1 www-data www-data   0  9 sept. 09:33 php8.4-fpm.sock
root@holdeck-serveur:~#
```

Les 2 sockets existent bien

Les 2 services FPM tournent

```
root@holdeck-serveur:~# systemctl status php8.4-fpm
● php8.4-fpm.service - The PHP 8.4 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.4-fpm.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-09 09:33:46 CEST; 6min ago
  Invocation: 1cc63c2b385242f0ae24ab45a397b57e
     Docs: man:php-fpm8.4(8)
    Main PID: 19164 (php-fpm8.4)
   Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
     Tasks: 3 (limit: 2255)
    Memory: 12.3M (peak: 14.3M)
       CPU: 131ms
   CGroup: /system.slice/php8.4-fpm.service
           └─19164 "php-fpm: master process (/etc/php/8.4/fpm/php-fpm.conf)"
             └─19168 "php-fpm: pool www"
               └─19169 "php-fpm: pool www"

sept. 09 09:33:46 holdeck-serveur systemd[1]: Starting php8.4-fpm.service - The PHP 8.4 FastCGI Process Manager...
sept. 09 09:33:46 holdeck-serveur systemd[1]: Started php8.4-fpm.service - The PHP 8.4 FastCGI Process Manager.
root@holdeck-serveur:~# systemctl status php7.4-fpm
● php7.4-fpm.service - The PHP 7.4 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php7.4-fpm.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-09 09:36:08 CEST; 4min 12s ago
  Invocation: fd43cba7b4a044cc8e87f73f706db08a
     Docs: man:php-fpm7.4(8)
    Process: 28772 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/7.4/fpm/pool.d/www.conf 74 (code=exited, status=0/SUCCESS)
    Main PID: 28767 (php-fpm7.4)
   Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"
     Tasks: 3 (limit: 2255)
    Memory: 10.8M (peak: 12.8M)
       CPU: 109ms
   CGroup: /system.slice/php7.4-fpm.service
           └─28767 "php-fpm: master process (/etc/php/7.4/fpm/php-fpm.conf)"
             └─28769 "php-fpm: pool www"
               └─28771 "php-fpm: pool www"

sept. 09 09:36:08 holdeck-serveur systemd[1]: Starting php7.4-fpm.service - The PHP 7.4 FastCGI Process Manager...
sept. 09 09:36:08 holdeck-serveur systemd[1]: Started php7.4-fpm.service - The PHP 7.4 FastCGI Process Manager.
lines 1-18/18 (END)
```

Services installés — VM Serveur

MariaDB	12.3.3	Dépôt officiel — sécurisée via mariadb-secure-installation
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Rajout clé + dépôt

```
root@holdeck-serveur:~# curl -Ls https://r.mariadb.com/downloads/mariadb_repo_setup | bash
# [info] Checking for script prerequisites.
# [info] MariaDB Server version 12.rolling is valid
# [info] Repository file successfully written to /etc/apt/sources.list.d/mariadb.sources
# [info] Adding trusted package signing keys...
# [info] Running apt-get update...
# [info] Done adding trusted package signing keys
root@holdeck-serveur:~#
```

```
root@holdeck-serveur:~# mariadb --version
mariadb from 12.3.3-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using EditLine wrapper
root@holdeck-serveur:~# systemctl status mariadb
● mariadb.service - MariaDB 12.3.3 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/mariadb.service.d
            └─migrated-from-my.cnf-settings.conf
   Active: active (running) since Wed 2026-09-09 10:08:18 CEST; 1min 13s ago
  Invocation: f5a06daf050840afbb195ea30c0600fc
     Docs: man:mariadb(8)
           https://mariadb.com/docs/server/server-management/starting-and-stopping-mariadb/systemd
   Process: 29770 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 29754 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 12 (limit: 14884)
   Memory: 122M (peak: 127.5M)
      CPU: 682ms
   CGroup: /system.slice/mariadb.service
           └─29754 /usr/sbin/mariabdd

sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] Plugin 'FEEDBACK' is disabled.
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] Plugin 'wsrep-provider' is disabled.
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] InnoDB: Buffer pool(s) load completed at 260909 10:08:18
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] mariabdd: Event Scheduler: Loaded 0 events
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: 2026-09-09 10:08:18 0 [Note] /usr/sbin/mariabdd: ready for connections.
sept. 09 10:08:18 holdeck-serveur mariabdd[29754]: Version: '12.3.3-MariaDB-deb13' socket: '/run/mysqld/mysqld.sock' port: 3306 mariadb.org binary distribu
sept. 09 10:08:18 holdeck-serveur systemd[1]: Started mariadb.service - MariaDB 12.3.3 database server.
sept. 09 10:08:18 holdeck-serveur /etc/mysql/debian-start[29772]: Upgrading MariaDB tables if necessary.
lines 1-27/27 (END)
```

```
Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
Query OK, 0 rows affected
SQL executed without errors!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
Query OK, 0 rows affected
SQL executed without errors!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
Query OK, 0 rows affected, 1 warning
SQL executed without errors!
- Removing privileges on test database...
Query OK, 0 rows affected
SQL executed without errors!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
root@holdeck-serveur:~#
```

Installation Maria DB effectué

MariaDB 12.3.3 installé depuis le dépôt officiel et démarré avec succès (service actif, en écoute sur le port 3306)

Services installés — VM Serveur

DHCP / DNS

isc-dhcp-server / bind9

Domaine starfleet.lan

```
root@holdeck-serveur:~# systemctl restart bind9
root@holdeck-serveur:~# systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-08 09:52:14 CEST; 19s ago
     Invocation: fd0ec1aed20f45c3ab166afcd3b14713
       Docs: man:named(8)
    Main PID: 6290 (named)
      Status: "running"
        Tasks: 8 (limit: 2255)
      Memory: 32.2M (peak: 34.4M)
         CPU: 41ms
    CGroup: /system.slice/named.service
            └─6290 /usr/sbin/named -f -u bind

sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 192.58.128.30#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './NS/IN': 192.58.128.30#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 192.36.148.17#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './NS/IN': 192.36.148.17#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 192.5.5.241#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './NS/IN': 192.5.5.241#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './DNSKEY/IN': 198.41.0.4#53
sept. 08 09:52:15 holdeck-serveur named[6290]: REFUSED unexpected RCODE resolving './NS/IN': 198.41.0.4#53
sept. 08 09:52:15 holdeck-serveur named[6290]: managed-keys-zone: Unable to fetch DNSKEY set '': failure
sept. 08 09:52:15 holdeck-serveur named[6290]: resolver priming query complete: failure
root@holdeck-serveur:~# _
```

Bind 9 tourne bien

```
GNU nano 8.4 /etc/bind/named.conf.local *
//
// Do any local configuration here
//

zone "starfleet.lan" {
    type master;
    file "/etc/bind/db.starfleet.lan";
};
```

on dit à bind9 "il existe un domaine starfleet.

```
root@holdeck-serveur:~# named-checkzone starfleet.lan /etc/bind/db.starfleet.lan
zone starfleet.lan/IN: loaded serial 1
OK
root@holdeck-serveur:~# named-checkconf
root@holdeck-serveur:~# _
```

Zone DNS validée

```
# }
# pool {
#     deny members of "foo";
#     range 10.0.29.10 10.0.29.230;
# }
# }
authoritative;

option domain-name "starfleet.lan";
option domain-name-servers 192.168.100.1;

subnet 192.168.100.0 netmask 255.255.255.0 {
    range 192.168.100.10 192.168.100.100;
    option routers 192.168.100.1;
    option subnet-mask 255.255.255.0;
    default-lease-time 600;
    max-lease-time 7200;
}

root@holdeck-serveur:~# dhcpd -t -f /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@holdeck-serveur:~# systemctl restart isc-dhcp-server
root@holdeck-serveur:~# systemctl status isc-dhcp-server
● isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
   Active: active (running) since Mon 2026-09-07 10:06:43 CEST; 7s ago
     Invocation: abbf9411329d4507bdfdaa39b46e44cf
       Docs: man:systemd-sysv-generator(8)
    Process: 2188 ExecStart=/etc/init.d/isc-dhcp-server start (code=exited, status=0/SUCCESS)
      Tasks: 1 (limit: 2255)
     Memory: 4.8M (peak: 6.5M)
        CPU: 54ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─2200 /usr/sbin/dhcpd -4 -q -cf /etc/dhcp/dhcpd.conf ens34

sept. 07 10:06:41 holdeck-serveur systemd[1]: Starting isc-dhcp-server.service - LSB: DHCP server...
sept. 07 10:06:41 holdeck-serveur isc-dhcp-server[2188]: Launching IPv4 server only.
sept. 07 10:06:41 holdeck-serveur dhcpd[2200]: Wrote 0 leases to leases file.
sept. 07 10:06:41 holdeck-serveur dhcpd[2200]: Server starting service.
sept. 07 10:06:43 holdeck-serveur isc-dhcp-server[2188]: Starting ISC DHCPv4 server: dhcpd.
sept. 07 10:06:43 holdeck-serveur systemd[1]: Started isc-dhcp-server.service - LSB: DHCP server.
root@holdeck-serveur:~#
```

DHCP tourne correctement

```
root@holdeck-serveur:~# dig @192.168.100.1 www8.starfleet.lan

; <<>> DiG 9.20.26-1~deb13u1-Debian <<>> @192.168.100.1 www8.starfleet.lan
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49258
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags::; udp: 1232
;; COOKIE: 6b2e9633a50f3ba3010000006a9fbff61281918236970086 (good)
;; QUESTION SECTION:
;www8.starfleet.lan.                IN      A

;; ANSWER SECTION:
www8.starfleet.lan.                604800  IN      A      192.168.100.1

;; Query time: 0 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Tue Sep 08 09:57:42 CEST 2026
;; MSG SIZE rcvd: 91

root@holdeck-serveur:~#
```

dig @192.168.100.1
www8.starfleet.lan zone
fonctionne

Services installés — VM Serveur

FTP

vsftpd

SSL/TLS obligatoire, chrooté sur /var/www

```
root@holdeck-serveur:/tmp# useradd -d /var/www -s /usr/sbin/nologin ftpuser
root@holdeck-serveur:/tmp# passwd ftpuser
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd : mot de passe mis à jour avec succès
root@holdeck-serveur:/tmp# _
```

créer un utilisateur FTP dédié (chrooté sur le dossier web)
apres avoir installé vsftpd

```
root@holdeck-serveur:/tmp# chown www-data:www-data /var/www
root@holdeck-serveur:/tmp# chmod 755 /var/www
root@holdeck-serveur:/tmp# _
```

donner les bons droits sur le dossier

```
GNU nano 8.4 /etc/vsftpd.conf *
listen=YES
listen_ipv6=NO

anonymous_enable=NO
local_enable=YES
write_enable=YES

chroot_local_user=YES
allow_writeable_chroot=YES

local_umask=022

pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40100

ssl_enable=YES
rsa_cert_file=/etc/ssl/starfleet/starfleet.crt
rsa_private_key_file=/etc/ssl/starfleet/starfleet.key
ssl_tlsv1=YES
ssl_sslv2=NO
ssl_sslv3=NO
force_local_data_ssl=YES
force_local_logins_ssl=YES

require_ssl_reuse=NO
ssl_ciphers=HIGH
```

sauvegarder la config par défaut et créer la nouvelle

```
GNU nano 8.4 /etc/nftables.conf *
#!/usr/sbin/nft -f

flush ruleset

table inet filter {
    chain input {
        type filter hook input priority 0; policy drop;

        #Autorise le trafic déjà initié par nous (réponse, etc.)
        ct state established,related accept

        # Autorise la boucle locale (indispensable, sinon le système casse)
        iif "lo" accept

        # Autorise le SSH pour moi (àdistance) - port 22
        tcp dport 22 accept
        udp dport 67 accept
        udp dport 53 accept
        tcp dport 53 accept
        tcp dport 80 accept
        tcp dport 443 accept
        tcp dport 21 accept
        tcp dport 40000-40100 accept
        # Autorise le ping (pratique pour tester, on pourra retirer plus tard)
        ip protocol icmp accept
    }

    chain forward {
        type filter hook forward priority 0; policy drop;
    }

    chain output {
        type filter hook output priority 0; policy accept;
    }
}
```

Rajout des ports FTP dan les parfeu

Services installés — VM Serveur

FTP

vsftpd

SSL/TLS obligatoire, chrooté sur /var/www

```
root@holdeck-serveur:/tmp# nft -f /etc/nftables.conf
root@holdeck-serveur:/tmp# nft list ruleset
table inet filter f {
    chain input {
        type filter hook input priority filter; policy drop;
        ct state established,related accept
        iif "lo" accept
        tcp dport 22 accept
        udp dport 67 accept
        udp dport 53 accept
        tcp dport 53 accept
        tcp dport 80 accept
        tcp dport 443 accept
        tcp dport 21 accept
        tcp dport 40000-40100 accept
        ip protocol icmp accept
    }

    chain forward {
        type filter hook forward priority filter; policy drop;
    }

    chain output {
        type filter hook output priority filter; policy accept;
    }
}
root@holdeck-serveur:/tmp# systemctl restart vsftpd
root@holdeck-serveur:/tmp# systemctl status vsftpd
• vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-10 15:25:33 CEST; 16s ago
   Invocation: 02fcd136db1343abaa1c28f4a9f6afb1
   Process: 37425 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 37428 (vsftpd)
   Tasks: 1 (limit: 2255)
   Memory: 1.5M (peak: 1.8M)
   CPU: 24ms
   CGroup: /system.slice/vsftpd.service
           └─37428 /usr/sbin/vsftpd /etc/vsftpd.conf

sept. 10 15:25:33 holdeck-serveur systemd[1]: Starting vsftpd.service - vsftpd FTP server...
sept. 10 15:25:33 holdeck-serveur systemd[1]: Started vsftpd.service - vsftpd FTP server.
root@holdeck-serveur:/tmp#
```

Le serveur FTP est configuré et actif

```
root@holdeck-serveur:/tmp# apt install lftp -y
Installation de :
  lftp

Sommaire :
  Mise à niveau de : 0, Installation de : 1Supprimé : 0, Non mis à jour : 0
Taille du téléchargement : 768 kB
Espace nécessaire : 2 434 kB / 28,4 GB disponible

Réception de : 1 http://deb.debian.org/debian trixie/main amd64 lftp amd64 4.9.2-3+b1 [768 kB]
768 ko réceptionnés en 0s (3 022 ko/s)
Sélection du paquet lftp précédemment désélectionné.
(Lecture de la base de données... 38389 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../lftp_4.9.2-3+b1_amd64.deb ...
Dépaquetage de lftp (4.9.2-3+b1) ...
Paramétrage de lftp (4.9.2-3+b1) ...
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
root@holdeck-serveur:/tmp# lftp -u ftpuser, ftp://192.168.100.1
lftp ftpuser@192.168.100.1:~> ls
ls: Erreur fatale: Certificate verification: The certificate is NOT trusted. The certificate issuer is unknown. (DF:0C:62:55:E4:53:66:B4:53:06:42:E3:4C:60:A2:A
A:48:3E:0A:7B)
lftp ftpuser@192.168.100.1:~> _
```

Test de connexion FTP en SSL TLS

```
root@holdeck-serveur:/tmp# cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/usr/bin/dash
/usr/sbin/nologin
root@holdeck-serveur:/tmp# exit
déconnexion
nouha@holdeck-serveur:~$ su -
Mot de passe :
root@holdeck-serveur:~# lftp -u ftpuser -e "set ssl:verify-certificate no" ftp://192.168.100.1
Mot de passe :
lftp ftpuser@192.168.100.1:~> ls
drwxrwxr-x   2 33      33          4096 Sep 10 08:50 _2admin
drwxrwxr-x  13 33      33          4096 Sep 10 08:24 phpmyadmin
drwxrwxr-x   2 33      33          4096 Sep 09 13:33 www7
drwxrwxr-x   2 33      33          4096 Sep 09 12:00 www8
lftp ftpuser@192.168.100.1:/>
```

rajout de nologin sur shell et fin de cette phase

Services installés — VM Serveur

LDAP	OpenLDAP (slapd)	dc=starfleet,dc=lan
------	------------------	---------------------

Ldap tourne bien 1

```
root@holdeck-serveur:~# systemctl status slapd
● slapd.service - OpenLDAP Server Daemon
  Loaded: loaded (/usr/lib/systemd/system/slapd.service; enabled; preset: enabled)
  Active: active (running) since Thu 2026-09-10 16:23:30 CEST; 53s ago
  Invocation: a2061ffe9648461381e43e718feb11f2
  Docs: man:slapd
        man:slapd-config
        man:slapd-mdb
  Main PID: 38295 (slapd)
  Tasks: 3 (limit: 2255)
  Memory: 5M (peak: 5.1M)
  CPU: 14ms
  CGroup: /system.slice/slapd.service
          └─38295 /usr/sbin/slapd -d0 -h "ldap:/// ldapi:///" -u openldap -g openldap

sept. 10 16:23:30 holdeck-serveur systemd[1]: Starting slapd.service - OpenLDAP Server Daemon...
sept. 10 16:23:30 holdeck-serveur slapd[38295]: @(#) $OpenLDAP: slapd 2.6.10+dfsg-1 (May 29 2025 23:41:48) $
                        Debian OpenLDAP Maintainers <pkg-openldap-devel@lists.alioth.debian.org>
sept. 10 16:23:30 holdeck-serveur slapd[38295]: slapd starting
sept. 10 16:23:30 holdeck-serveur systemd[1]: Started slapd.service - OpenLDAP Server Daemon.
root@holdeck-serveur:~# _
```

créer un dossier LDIF 2

```
GNU nano 8.4 /root/structure.ldif *
dn: ou=people,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: people

dn: ou=groups,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: groups_
```

```
root@holdeck-serveur:~# slapcat | head -20
dn: dc=starfleet,dc=lan
objectClass: top
objectClass: dcObject
objectClass: organization
o: starfleet.lan
dc: starfleet
structuralObjectClass: organization
entryUUID: f0ccbf78-416e-1041-919a-8fd5948ee693
creatorsName: cn=admin,dc=starfleet,dc=lan
createTimestamp: 20260910142330Z
entryCSN: 20260910142330.335018Z#000000#000#000000
modifiersName: cn=admin,dc=starfleet,dc=lan
modifyTimestamp: 20260910142330Z

root@holdeck-serveur:~# _
```

créer un dossier LDIF 3

```
objectClass: shadowAccount
uid: kirk
sn: kirk
givenName: James
cn: James Kirk
displayName: James Kirk
uidNumber: 10000
gidNumber: 10000
userPassword: {SSHA}R/CrETKb197wrE+iNo2EJI+sLVVH5XKt
gecos: James Kirk
loginShell: /bin/bash
homeDirectory: /home/kirk

root@holdeck-serveur:~# ldapadd -x -D "cn=admin,dc=starfleet,dc=lan" -H -f /root/utilisateur.ldif
adding new entry "uid=kirk,ou=people,dc=starfleet,dc=lan"

root@holdeck-serveur:~# _
```

L'utilisateur kirk est bien créé dans ton annuaire LDAP. 5

```
root@holdeck-serveur:~# slappasswd
New password:
Re-enter new password:
{SSHA}R/CrETKb197wrE+iNo2EJI+sLVVH5XKt
root@holdeck-serveur:~# _
```

Générer un mot de passe 4

Les 4 sites web

www8.starfleet.lan

Site web — PHP 8.4

www7.starfleet.lan

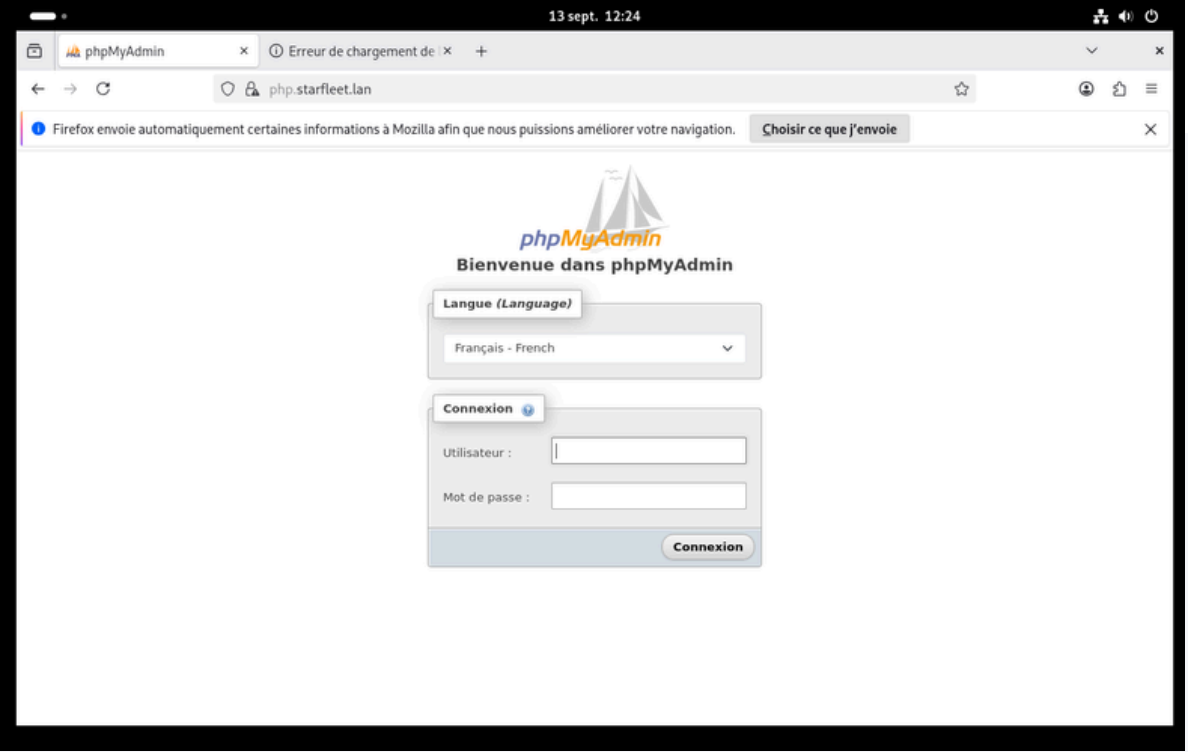
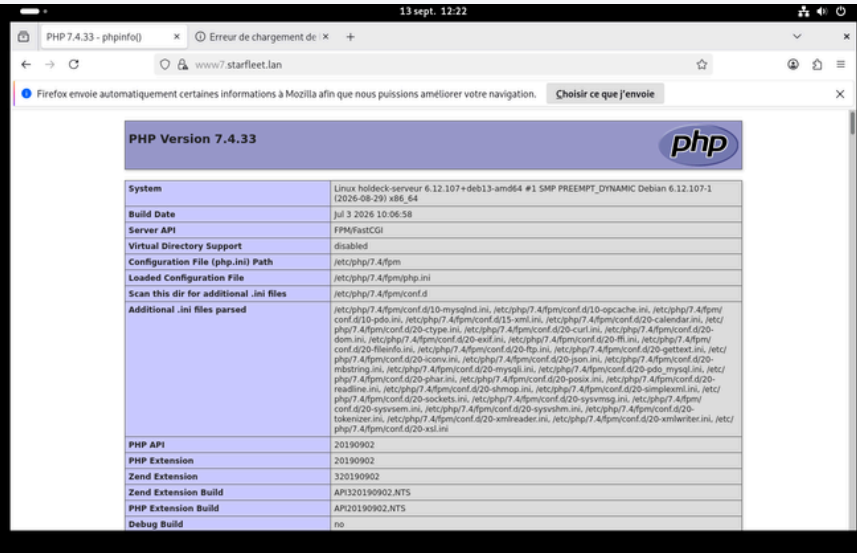
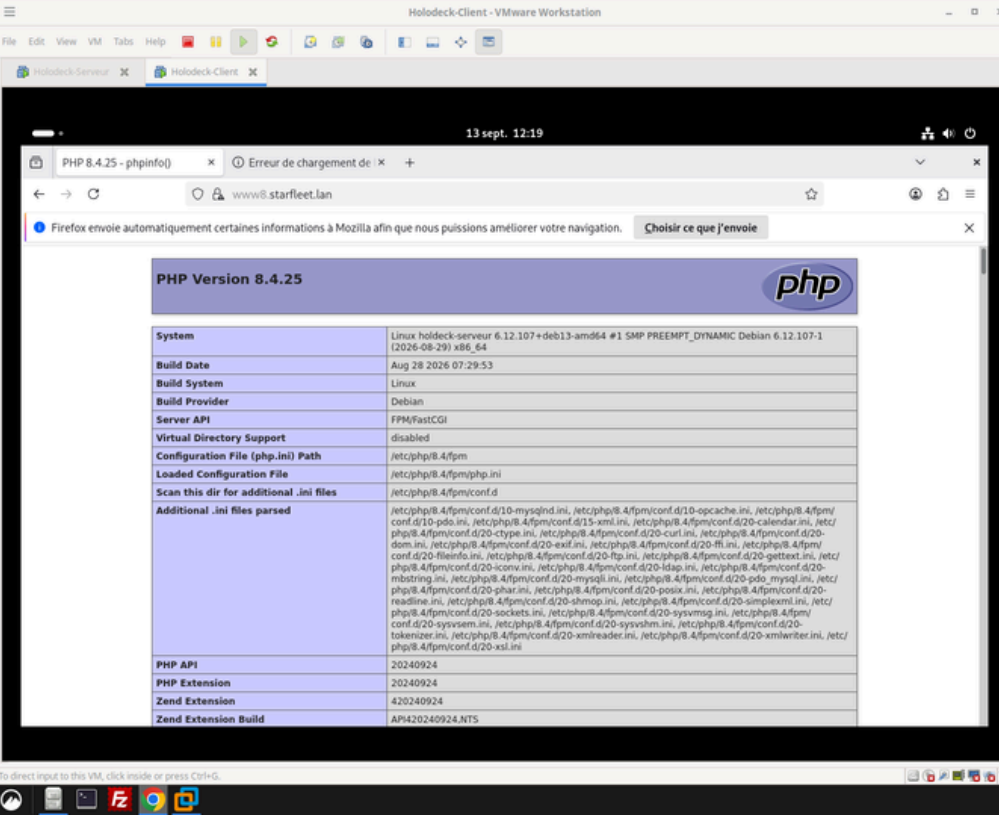
Site web — PHP 7.4

php.starfleet.lan

phpMyAdmin

_2admin.starfleet.lan

Administration de la VM



Les 4 sites web



Probablement un beuge, impossible de
mettre le code

```
root@holdeck-serveur:~# curl -k https://www8.starfleet.lan/ldap_login.php | grep password
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100    258      0   258    0     0   18767      0 --:--:-- --:--:-- --:--:-- 19846
Mot de passe : <input type="password" name="password"><br>
root@holdeck-serveur:~# curl -k -X POST -d "uid=kirk&password=x" https://www8.starfleet.lan/ldap_login.php
<html>
<body>
<h1>Connexion via annuaire LDAP</h1>
<p style='color:green;'>Connexion reussie pour kirk !</p><form method="post">
  Identifiant / <input type="text" name="uid"><br>
  Mot de passe : <input type="password" name="password"><br>
  <input type="submit" value="Se connecter">
</form>
</body>
</html>
root@holdeck-serveur:~# curl -k -X POST -d "uid=kirk&password=x" https://www8.starfleet.lan/ldap_login.php
<html>
<body>
<h1>Connexion via annuaire LDAP</h1>
<p style='color:green;'>Connexion reussie pour kirk !</p><form method="post">
  Identifiant / <input type="text" name="uid"><br>
  Mot de passe : <input type="password" name="password"><br>
  <input type="submit" value="Se connecter">
</form>
</body>
</html>
root@holdeck-serveur:~# curl -k -X POST -d "uid=kirk&password=s" https://www8.starfleet.lan/ldap_login.php
<html>
<body>
<h1>Connexion via annuaire LDAP</h1>
<p style='color:red;'>Echec de connexion. Identifiants
incorrects.</p><form method="post">
  Identifiant / <input type="text" name="uid"><br>
  Mot de passe : <input type="password" name="password"><br>
  <input type="submit" value="Se connecter">
</form>
</body>
</html>
root@holdeck-serveur:~#
```

authentification ldap testée (kirk) qui est
validée via curl.

Bilan et perspectives

- Infrastructure complète validée : DHCP, DNS, 4 sites HTTPS, FTP sécurisé, LDAP
- Reste à faire : export des VM, dépôt GitHub public
- Bonus envisageable : VS Code Server, script de sauvegarde automatisée

Merci de votre attention